

Reasoning about Bias in MASs Verification

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Outline

- 1 Motivations and overview
- 2 Preliminaries: CGS and ATL
- 3 Modelling Bias
- 4 The Bias-Sensitive Logic
- 5 Model Checking bATL and bATL*
- 6 Conclusion and Future Works

Reasoning about bias/fairness

Motivations

- Bias in *decision-making* systems is a growing concern
 - ▶ intelligent systems increasingly support or automate decisions in e.g., hiring, reviewing, justice, healthcare,
- Bias in MASs is *subtle* and *dynamic*
 - ▶ bias may arise not only from individual agents but from strategic behaviours over time.
- Existing approaches are limited, struggle to capture:
 - ▶ strategic behaviour,
 - ▶ temporal evolution, and
 - ▶ interaction-driven bias.

Reasoning about bias/fairness

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Aim

Develop formal framework to reason about **bias** in dynamic MASs.

Reasoning about Bias: Key ideas and Contributions

Fairness as non-interference

We introduce a novel definition of bias inspired by **non-interference**, intuitively:

- A system is unbiased if **bias-sensitive attributes** (e.g., gender, race) do not affect behaviour or outcomes when non-bias attributes are fixed.
- Captured via **state and path equivalence**.

Why this matters?

- Goes beyond simple outcome comparison.
- Captures implicit, structural, and interaction-driven bias.
- Well-suited to strategic MASs.

Key ideas and Contributions

Bias-aware temporal logic

- We propose bATL and bATL*, extensions of ATL/ATL*, with bias-sensitive operators.
- Enable formal specification of fairness and bias properties under strategies.

Verification guarantees

- We develop model-checking algorithms for both logics
- Complexity results:
 - ▶ bATL: PTIME-complete
 - ▶ bATL*: 2EXPTIME-complete

Contribution

A unified logical and algorithmic framework for **formal bias reasoning in strategic, temporal MASs**.

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Concurrent Game Structure (CGS)

CGS $M = \langle \text{Ag}, \text{Ap}, S, s_I, \{\text{Act}_i\}_{i \in \text{Ag}}, d, \delta, V \rangle$ such that:

- $\text{Ag} = \{1, \dots, m\}$ is a nonempty finite set of agents.
- Ap is a nonempty finite set of atomic propositions (atoms).
- $S \neq \emptyset$ is a finite set of *states*, with a set of *initial states* $s_I \subseteq S$.
- For $i \in \text{Ag}$, Act_i is a finite set of *actions*, $\text{Act} = \bigcup_{i \in \text{Ag}} \text{Act}_i$, and $\text{ACT} = \prod_{i \in \text{Ag}} \text{Act}_i$ the set of all joint actions.
- The *protocol function* $d : \text{Ag} \times S \rightarrow (2^{\text{Act}} \setminus \{\emptyset\})$
- The *transition function* $\delta : S \times \text{ACT} \rightarrow S$ assigns a successor state $s' = \delta(s, \vec{a})$ to each $s \in S$, for every joint action $\vec{a} \in \text{ACT}$ such that $a_i \in d(i, s)$ for every $i \in \text{Ag}$.
- $V : S \rightarrow 2^{\text{Ap}}$ is the *labelling function*.

Preliminaries

Syntax: ATL/ATL*

Formulas in ATL*:

$$\varphi ::= q \mid \neg\varphi \mid \varphi \wedge \varphi \mid \langle\langle\Gamma\rangle\rangle\psi$$

$$\psi ::= \varphi \mid \neg\psi \mid \psi \wedge \psi \mid \bigcirc\psi \mid (\psi \mathbf{U} \psi)$$

Formulas in the ATL fragment are obtained from defn of ATL*:

$$\psi ::= \bigcirc\varphi \mid (\varphi \mathbf{U} \varphi) \mid (\varphi \mathbf{R} \varphi)$$

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Modelling Bias

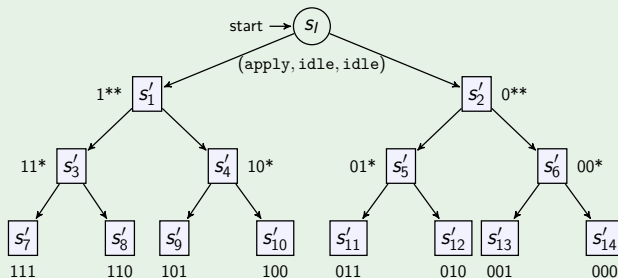
Bias-sensitive Model

A **bias-sensitive model** is a tuple of the form $\mathcal{G}=(M, \eta)$, where:

- M is a concurrent game structure (CGS);
- $\eta : \text{Ap} \mapsto \mathcal{L}$ is the bias labelling function mapping each atomic proposition to a bias level in $\mathcal{L}$ representing the complete lattice of the bias levels.

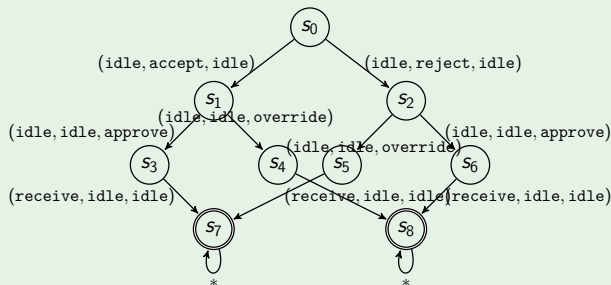
Modelling Bias

Example: college admission system



Modelling Bias

Example: college admission system



$s_0 = \{s'_i \mid 7 \leq i \leq 12\}$: an application received for evaluation.

s_1 and s_2 : AI reviewer recommending "accept" and "reject".

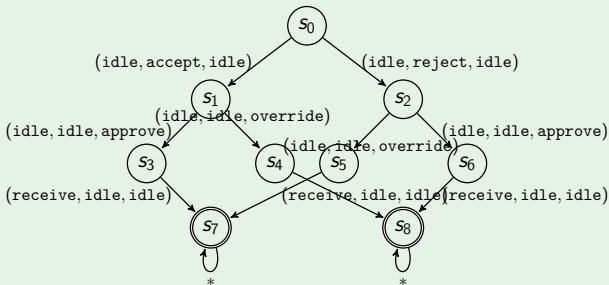
s_3 and s_6 : human reviewer confirming that of AI reviewer.

s_4 and s_5 : human reviewer "overrides" AI reviewer's recommendation.

s_7 and s_8 : the final admission outcomes "accept" and "reject".

Modelling Bias

Example: college admission system



Example paths with input merit = 1, origin = 1, group = 0:

$$\pi_1 = (s'_8 \in s_0) \rightarrow s_2 \rightarrow s_6 \rightarrow s_8 \quad \pi_2 = (s'_8 \in s_0) \rightarrow s_2 \rightarrow s_5 \rightarrow s_7$$

and example paths with input merit = 1, origin = 1, group = 1:

$$\pi_3 = (s'_7 \in s_0) \rightarrow s_1 \rightarrow s_3 \rightarrow s_7 \quad \pi_4 = (s'_7 \in s_0) \rightarrow s_1 \rightarrow s_4 \rightarrow s_8$$

Formalising Bias

Bias levels

- We assign bias-sensitive levels with ordering to state atomic propositions, through $\eta : \text{Ap} \mapsto \mathcal{L}$, where $\mathcal{L}$ denotes the complete lattice of the bias-sensitive levels.
- The powerset of atomic state propositions therefore forms a complete lattice $\mathcal{L}$, whose partial ordering is regarding to the bias-sensitive levels of the atomic propositions Ap :

$$\forall v, v' \in \text{Ap}, \quad v \preceq v' \text{ iff } \eta(v) \leq \eta(v')$$

Formalising Bias

Non-Bias Equivalent States

We say state $s, s' \in S$ are $\mathcal{NB}$ -equivalent, written as $s =_{\mathcal{NB}} s'$, iff:

$$\forall x \in \text{Ap}. \eta(x) \leq \mathcal{NB} \Rightarrow (x \in V(s) \wedge x \in V(s')) \\ \vee (x \notin V(s) \wedge x \notin V(s')).$$

Example: NB-equivalent states

let $\eta(\text{merit}) = \eta(\text{origin}) = \eta(\text{decision}) = \mathcal{NB}$, and $\eta(\text{group}) = \mathcal{B}$, and $\mathcal{NB} \preceq \mathcal{B}$, i.e., demographic group is bias-sensitive attribute while merit, origin, and final decision are non-bias-sensitive. We have:

$$s'_7 =_{\mathcal{NB}} s'_8, \quad s'_9 =_{\mathcal{NB}} s'_{10}, \quad s'_{11} =_{\mathcal{NB}} s'_{12}, \quad s'_{13} =_{\mathcal{NB}} s'_{14}.$$

Formalising Bias

Non-Bias Equivalent Paths

Given a model $\mathcal{G}$, and two paths $\pi, \pi' \in \text{Path}_{\mathcal{G}}$. We say that two paths π and π' are **weakly $\mathcal{NB}$ -equivalent**, written as $\pi \sim_{\mathcal{NB}_w} \pi'$, iff:

- If $|\pi| < |\pi'|$: $\pi_0 =_{\mathcal{NB}} \pi'_0 \Rightarrow \exists j \geq |\pi|. \forall k \geq j. (last(\pi) =_{\mathcal{NB}} \pi'_k)$;
- If $|\pi'| < |\pi|$: $\pi_0 =_{\mathcal{NB}} \pi'_0 \Rightarrow \exists j \geq |\pi'|. \forall k \geq j. (\pi_k =_{\mathcal{NB}} last(\pi'))$;
- Else $|\pi| = |\pi'|$: $\pi_0 =_{\mathcal{NB}} \pi'_0 \Rightarrow last(\pi) =_{\mathcal{NB}} last(\pi')$.

π and π' are **strongly $\mathcal{NB}$ -equivalent**, written as $\pi \sim_{\mathcal{NB}_s} \pi'$, iff:

$$(|\pi| = |\pi'|) \wedge (\pi_0 =_{\mathcal{NB}} \pi'_0 \Rightarrow \forall j \in (0, |\pi|]. (\pi_j =_{\mathcal{NB}} \pi'_j)).$$

Formalising Bias

Example: NB-equivalent paths

Consider example paths with input merit = 1, origin = 1, group = 0:

$$\pi_1 = (s'_8 \in s_0) \rightarrow s_2 \rightarrow s_6 \rightarrow s_8$$

$$\pi_2 = (s'_8 \in s_0) \rightarrow s_2 \rightarrow s_5 \rightarrow s_7$$

and example paths with input merit = 1, origin = 1, group = 1:

$$\pi_3 = (s'_7 \in s_0) \rightarrow s_1 \rightarrow s_3 \rightarrow s_7$$

$$\pi_4 = (s'_7 \in s_0) \rightarrow s_1 \rightarrow s_4 \rightarrow s_8$$

Clearly: $\pi_1 \not\sim_{\mathcal{NB}_w} \pi_3$, $\pi_1 \sim_{\mathcal{NB}_w} \pi_4$, $\pi_2 \sim_{\mathcal{NB}_w} \pi_3$, $\pi_2 \not\sim_{\mathcal{NB}_w} \pi_4$.

Formalising Bias

Bias Plicy

Let ψ be an outcome formula and $\mathcal{G} = (\text{Ag}, S, s_I, \text{Act}, \rightarrow, \text{Ap}, L, \eta)$ a model.

- The model $\mathcal{G}$ is said to be **weakly biased** with respect to ψ , written $\mathcal{G} \Vdash_{\psi} \phi_{\mathcal{B}_w}$, iff

$$\text{Paths}_{\mathcal{G}}(s_I, \psi) = \emptyset \quad \vee \quad \exists \pi, \pi' \in \text{Paths}_{\mathcal{G}}(s_I, \psi) \text{ implies } \pi \not\sim_{\mathcal{N}\mathcal{B}_w} \pi'.$$

- The model $\mathcal{G}$ is said to be **strongly biased** with respect to ψ , written $\mathcal{G} \Vdash_{\psi} \phi_{\mathcal{B}_s}$, iff

$$\text{Paths}_{\mathcal{G}}(s_I, \psi) = \emptyset \quad \vee \quad \exists \pi, \pi' \in \text{Paths}_{\mathcal{G}}(s_I, \psi) \text{ implies } \pi \not\sim_{\mathcal{N}\mathcal{B}_s} \pi'.$$

Formalising Bias

Example: Bias Plicy

- Let $\psi = \Diamond(\text{decision})$, note that:
- there exists $\pi_1, \pi_3 \in \text{Paths}_{\mathcal{G}}(s_I, \psi)$ s.t. $\pi_1 \not\sim_{\mathcal{NB}_w} \pi_3$, so:
 - ▶ $\mathcal{G}$ is a weakly biased model regarding ψ .
- This meets our intuition since:
 - ▶ the bias-sensitive attribute of demographic group impacts the final decision while non-bias-sensitive attributes remains the same, i.e.,
 - ▶ the final result is influenced by bias-sensitive attributes, throughout the path.

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The Bias-Sensitive Logic: bATL*

Syntax: bATL*

$$\begin{aligned}\varphi &::= q \mid \neg\varphi \mid \varphi \wedge \varphi \mid \langle\!\langle\Gamma\rangle\!\rangle\psi \mid \langle\!\langle\Gamma\rangle\!\rangle\mathfrak{B}_w[\psi] \mid \langle\!\langle\Gamma\rangle\!\rangle\mathfrak{B}_s[\psi] \\ \psi &::= \varphi \mid \neg\psi \mid \psi \wedge \psi \mid \bigcirc\psi \mid (\psi \mathbf{U} \psi)\end{aligned}$$

Example

- $\neg\langle\!\langle A \rangle\!\rangle\mathfrak{B}_w[\Diamond(\text{policyPassed})]$ captures that bias-sensitive factors (e.g., media influence, institutional discrimination) prevent party A from ensuring the policy's success through any strategy;
- $\langle\!\langle\Gamma\rangle\!\rangle\mathfrak{B}_w[\Box \neg\text{error}]$ states that under weak bias consideration, coalition Γ can ensure that no execution path leads to an error state.

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Model Checking bATL

Verification problem for bATL formulae

Given a bias-sensitive model $\mathcal{G}=(\text{Ag}, \text{Ap}, S, s_I, \{\text{Act}_i\}_{i \in \text{Ag}} \delta, V, \eta)$ and a bATL formula φ , determine whether it is the case that every initial state $s \in s_I$ satisfies φ .

Theorem

The model checking problem for bATL in our model is decidable, and can be done in PTIME-complete.

Theorem

The model checking problem for bATL^{*} in CGSs is decidable, and can be done in 2EXPTIME-complete.

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Conclusion and Future Works

Summary

- A novel notion of bias and a corresponding bias policy for MASs,
- bATL logic framework upon a game structure model.
- Verification process for reasoning about bias within this framework.

Future works

- Extend our framework to incorporate quantitative measures of bias.
- Investigate how strategic interactions among agents affect bias.
- Integrating strategic elements into our logical framework to better capture the complexities of decision-making in competitive and cooperative environments.